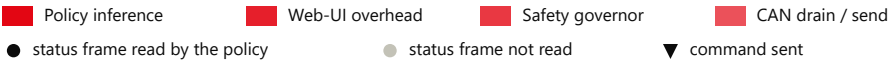
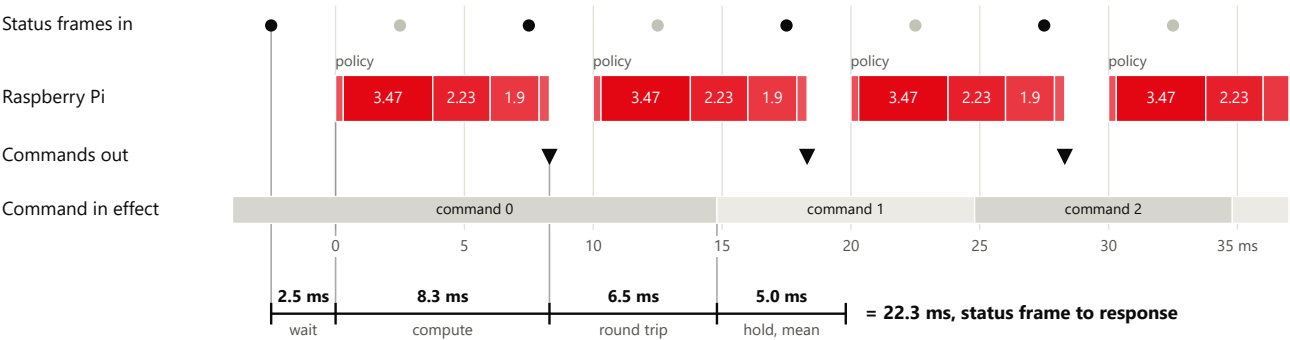


Control-loop timeline

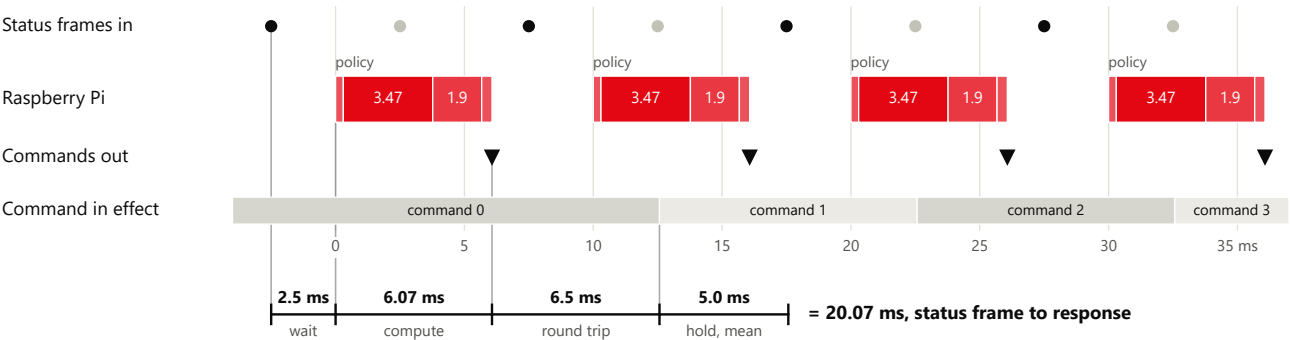
Status frame in, policy, command out, response — robot's Raspberry Pi 3B at a nominal 100 Hz



Web-UI daemon: 10 ms loop slots. Policy tick 8.3 ms, policy period 10.0 ms (100 Hz)



Headless run_policy.py, web UI stopped: 10 ms loop slots. Policy tick 6.07 ms, policy period 10.0 ms (100 Hz)



Tick durations: CAN drain 0.3 ms (inferred), policy inference 3.47 ms alone, web-UI overhead 2.23 ms (5.3–6.1 ms in the daemon, midpoint, drawn as one block), safety governor 1.9 ms, CAN send 0.4 ms. Loop rule (daemon.py_loop, run_policy.py): sleep until next_t += slot; after an overrun, next_t restarts from the end of the tick.

Status frames: each drive every 5 ms, not synchronised to the loop; mean wait 2.5 ms. Round trip 6.5 ms: command frame out to response visible in a status frame (MIT sweep) = drive command path + motor response + status path. Hold: a command stays in effect for one policy period; mean age half a period.